

TMF1414 Introduction to Programming
Pair Assignment (10%) – (2 students in a group)
Due Date: 1st December 2025, before 4pm

Learning Outcomes:

To demonstrate the ability of students to:

- Distinguish the basic problem-solving techniques in developing algorithms and programs for given problems. [C4]
- Construct complete programs based on a given specification. [P4]
- Construct the process of top-down, stepwise refinement to benefit all related software construction task. [P4]

Objectives:

To demonstrate the ability of students to:

- analyse given problem and identify correct requirements.
- produce readable program – indentation, comments, variables
- produce a correct programming code based on programming design
- design solution and use appropriate control structures to handle program logics based on different conditions and generate correct outputs. For instance, selection and repetition, multiple selections and others control statements.

Problem Statement(s):

Write a program for **PawPal Pet Centre** to compute grooming and boarding charges for pets based on service type and number of days stayed.

Requirements:

- Parking charge table:

Service Type	Base Grooming Fee
Basic Wash	35.00
Full Grooming	60.00
Spa & Groom	90.00

- Boarding Charges: RM20 per pet per day
- Discount based on total days:
 - If <5 days → 0% on boarding (no discount)
 - If 5–9 days → 5% on boarding
 - If ≥10 days → 10% on boarding
- Repeat until sentinel (Y/N)
- Validate number of days is ≥ 0 (i.e., not a negative value)
- For each pet calculate:
 - Grooming Fee
 - Boarding Fee = days $\times$ RM20 minus discount (if any)

- Total = Grooming + Boarding
- Daily summary must include:
 - Total pets serviced
 - Total grooming fees
 - Total boarding fees
 - Grand total

Example Output:

Welcome to PawPal Service!

Pet name: Mochi

Service (1=Basic, 2=Full, 3=Spa): 3

Boarding days: 7

Grooming Fee: RM90.00

Boarding Fee (after discount): RM133.00

Total: RM223.00

Next pet? (Y/N): Y

...

...

Process next? (Y/N): N

****Daily summary****

Total Pets: 4

Total Grooming Fees: RM240.00

Total Boarding Fees: RM360.00

Grand Total: RM600.00

Deliverables:

- a) A set of bug-free program (source code) - sc_STUDENTIDs.c (eg: sc_10021_10033.c).
- b) Refer to the assignment rubric on how to document your report (eg: rpt_10021_10033.docx). Your report should not exceed 5 pages (excluding the cover and the table of content). Font type is Century Schoolbook and font size is 11 with single spacing. Your report should contain;
 - i. Assignment cover
 - ii. Table of content
 - iii. Introduction
 - iv. Flow charts
 - v. Pseudocode
 - vi. Screenshots (sample run of your program) with explanation.
 - vii. Proof of discussion (WhatsApp or meeting record/pics).

Remarks:

- This assignment must be done in **pair (2 students)**;
- Plagiarism will be penalized; no mark will be given;
- The assignment must be submitted through **eleap.unimas.my** (C source code, and softcopy of the report) as one zip file and name it with your matric number, for example “10021_10033.zip”. You must upload it to **eleap.unimas.my** by **1st December 2025, before 4.00pm**; and
- Late submission will only receive up to 50% out of the total mark.